

SOAL 2 Enkripsi Citra logo Toko dengan Invers matriks 3x3

Parameter Personalisasi

$$N = 10, a = 1, b = 0, k = 2, \theta = 30^\circ, \alpha = 0,001$$

➤ Matriks kunci

$$A = \begin{vmatrix} a+1 & b & 2 \\ \cdot 1 & a+2 & \cdot b \\ \cdot b & 1 & a+3 \end{vmatrix}$$

➤ substitusi nilai a dan b

$$A = \begin{vmatrix} 1+1 & 0 & \cdot 2 \\ 1 & 1+2 & 0 \\ 0 & 1 & 1+3 \end{vmatrix}$$

a) matriks A

$$A = \begin{vmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 1 & 4 \end{vmatrix}$$

Determinan A

Ekspansi kefaktor baris pertama

$$A = \begin{vmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 1 & 4 \end{vmatrix}$$

$$\det(A) = 2 \cdot \begin{vmatrix} 3 & 0 \\ 1 & 4 \end{vmatrix} - 0 \cdot \begin{vmatrix} 1 & 0 \\ 0 & 4 \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix}$$

$$= 2 \cdot ((3 \cdot 4) - 0) - 0 + 2 \cdot ((1 \cdot 1) - 0)$$

$$= 2 \cdot 12 + 2$$

$$= 24 + 2$$

$$= 26$$

$$\boxed{\det(A) = 26}$$

b) matriks minor

$$A = \begin{vmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 1 & 4 \end{vmatrix}$$

M _{ij}	matriks	Nilai
M ₁₁	$\begin{vmatrix} 3 & 0 \\ 1 & 4 \end{vmatrix}$	12
M ₁₂	$\begin{vmatrix} 1 & 0 \\ 0 & 4 \end{vmatrix}$	4
M ₁₃	$\begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix}$	1
M ₂₁	$\begin{vmatrix} 0 & 2 \\ 1 & 4 \end{vmatrix}$	-2
M ₂₂	$\begin{vmatrix} 2 & 2 \\ 0 & 4 \end{vmatrix}$	8
M ₂₃	$\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix}$	2
M ₃₁	$\begin{vmatrix} 0 & 2 \\ 3 & 0 \end{vmatrix}$	-6
M ₃₂	$\begin{vmatrix} 2 & 2 \\ 1 & 0 \end{vmatrix}$	-2
M ₃₃	$\begin{vmatrix} 2 & 0 \\ 1 & 3 \end{vmatrix}$	6

3) Invers matriks

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{Adj}(A)$$

$$A^{-1} = \frac{1}{26} \cdot \begin{vmatrix} 12 & 2 & -6 \\ -4 & 8 & 2 \\ 1 & -2 & 6 \end{vmatrix}$$

$$A^{-1} = \begin{vmatrix} \frac{12}{26} & \frac{2}{26} & \frac{-6}{26} \\ \frac{-4}{26} & \frac{8}{26} & \frac{2}{26} \\ \frac{1}{26} & \frac{-2}{26} & \frac{6}{26} \end{vmatrix}$$

d) matlab

e) Blok Pixel (3 x 3)

$$P = \begin{vmatrix} 120 & (50+N) & 200 \\ 80 & (1+2) & (10+N) \\ (30+N) & 90 & 170 \end{vmatrix}$$

$$P = \begin{vmatrix} 120 & (50+10) & 200 \\ 80 & (1+2) & (10+10) \\ (30+10) & 90 & 170 \end{vmatrix}$$

c) kofaktor, Adjoin, dan Invers

1) matriks kofaktor C

$$C = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} \begin{vmatrix} 12 & -4 & 1 \\ 2 & 8 & -2 \\ -6 & 2 & 6 \end{vmatrix}$$

$$P = \begin{vmatrix} 120 & 60 & 200 \\ 80 & 3 & 20 \\ 40 & 90 & 170 \end{vmatrix}$$

2) Adj A = Transpose kofaktor

$$\text{Adj } A = C^T = \begin{vmatrix} 12 & 2 & -6 \\ -4 & 8 & 2 \\ 1 & -2 & 6 \end{vmatrix}$$

Matriks A

$$A = \begin{vmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 1 & 4 \end{vmatrix}$$

1) Enkripsi $E = A \cdot P$

Baris 1 :

$$E_{11} = 2(120) + 0(80) + 2(40) = 240 + 0 + 80 = 320$$

$$E_{12} = 2(60) + 0(3) + 2(90) = 120 + 0 + 180 = 300$$

$$E_{13} = 2(200) + 0(20) + 2(170) = 400 + 0 + 340 = 740$$

Baris 2 :

$$E_{21} = 1(120) + 3(80) + 0(40) = 120 + 240 + 0 = 360$$

$$E_{22} = 1(60) + 3(3) + 0(90) = 60 + 9 + 0 = 69$$

$$E_{23} = 1(200) + 3(20) + 0(170) = 200 + 60 + 0 = 260$$

Baris 3 :

$$E_{31} = 0(120) + 1(80) + 4(40) = 0 + 80 + 160 = 240$$

$$E_{32} = 0(60) + 1(3) + 4(90) = 0 + 3 + 360 = 363$$

$$E_{33} = 0(200) + 1(20) + 4(170) = 0 + 20 + 680 = 700$$

Hasil Enkripsi

$$E = \begin{vmatrix} 320 & 300 & 740 \\ 360 & 69 & 260 \\ 240 & 363 & 700 \end{vmatrix}$$

2) Dekripsi $P' = A^{-1} \cdot E$

$$A^{-1} = \frac{1}{26} \cdot \begin{vmatrix} 12 & 2 & -6 \\ -4 & 8 & 2 \\ 1 & -2 & 6 \end{vmatrix}$$

$$E = \begin{vmatrix} 320 & 300 & 740 \\ 360 & 69 & 260 \\ 240 & 363 & 700 \end{vmatrix}$$

Baris 1

D/ P'_{11}

$$= \frac{1}{26} (12 \cdot 320 + 2 \cdot 360 - 6 \cdot 240)$$

$$= \frac{1}{26} (3840 + 720 - 1440)$$

$$= \frac{3120}{26}$$

$$= 120$$

▷ P'12

$$= \frac{1}{26} (12 \cdot 300 + 2 \cdot 69 - 6 \cdot 363)$$

$$= \frac{1}{26} (3600 + 138 - 2178)$$

$$= \frac{1560}{26}$$

$$= 60$$

▷ P'13

$$= \frac{1}{26} (12 \cdot 740 + 2 \cdot 260 - 6 \cdot 700)$$

$$= \frac{1}{26} (8880 + 520 - 4200)$$

$$= \frac{5200}{26}$$

$$= 200$$

Baris 2

▷ P'21

$$= \frac{1}{26} (-4 \cdot 320 + 8 \cdot 360 + 2 \cdot 240)$$

$$= \frac{1}{26} (-1280 + 2880 + 480)$$

$$= \frac{2080}{26}$$

$$= 80 //$$

▷ P'22

$$= \frac{1}{26} (-4 \cdot 300 + 8 \cdot 69 + 2 \cdot 363)$$

$$= \frac{1}{26} (-1200 + 552 + 726)$$

$$= \frac{78}{26}$$

$$= 3 //$$

▷ P'23

$$= \frac{1}{26} (-4 \cdot 740 + 8 \cdot 260 + 2 \cdot 700)$$

$$= \frac{1}{26} (-2960 + 2080 + 1400)$$

$$= \frac{520}{26}$$

$$= 20 //$$

Baris 3

▷ P'31

$$= \frac{1}{26} (1 \cdot 320 - 2 \cdot 360 + 6 \cdot 240)$$

$$= \frac{1}{26} (320 - 720 + 1440)$$

$$= \frac{1040}{26}$$

$$= 40$$

▷ P'32

$$= \frac{1}{26} (1 \cdot 300 - 2 \cdot 69 + 6 \cdot 363)$$

$$= \frac{1}{26} (300 - 138 + 2178)$$

$$= \frac{2340}{26}$$

$$= 90$$

▷ P'33

$$= \frac{1}{26} (1 \cdot 740 - 2 \cdot 260 + 6 \cdot 700)$$

$$= \frac{1}{26} (740 - 520 + 4200)$$

$$= \frac{4420}{26}$$

$$= 170$$

Hasil akhir :

	120	60	200
P' =	80	3	20
	40	90	170

	120	60	200
P =	80	3	20
	40	90	170

$$\therefore P' = P$$

SiDU